
Tools, Not Talk: Biology Agents Hit a Knowledge Ceiling That Orchestration and Reasoning Cannot Break

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Abstract

Agentic systems for biology are increasingly built around orchestration: critics that review answers, teams of role-playing scientists that debate, and models that reason for longer. We show that on objectively scored biology research tasks this machinery buys almost nothing, and we identify why. Across 150 pre-registered LAB-Bench and LABBench2 questions, compared at a matched budget of three model calls, self-critique never beats simple majority voting, multi-agent debate helps only a model whose reasoning is switched off, and a 25-fold increase in generated tokens from longer reasoning leaves accuracy unchanged. The reason is a ceiling set by the questions themselves: which question is asked explains 56% of the variance in correctness, while the choice of model, strategy and reasoning effort explains 0.6%. That ceiling is missing knowledge, not missing deliberation. A single model call with web search and code execution solves 47% of the questions that every orchestrated and reasoning-heavy system failed. For life-science agents, the budget belongs in tools, not talk.

1 Introduction

LLM agents now plan experiments, analyse data and generate hypotheses in chemistry and biology [1, 2, 5, 6, 8, 16]. Many of these systems are organised as conversations: a critic reviews a draft, several specialised scientists debate, an adjudicator decides. Their gains are usually reported against a single model call, even though the agent makes many. Sampling several answers and voting already improves accuracy [3, 12, 19], so these comparisons cannot tell whether the orchestration helps or whether the extra calls do [9].

We separate the two. Every orchestrated strategy receives exactly the budget of a voting baseline: three calls, the same model, the same prompts and the same decoding. We then ask what actually moves accuracy on biology research tasks: orchestration, longer reasoning, or access to knowledge through tools.

Contributions.

- **Orchestration buys little at matched compute.** Across three models, self-critique never beats voting and repairs no more answers than it breaks. Debate helps only when the model cannot reason on its own. Raising reasoning effort, for six model families, rarely helps either (§3.1–3.2).
- **The question decides.** Every system solves the same questions and fails the same questions. Question identity explains 56% of the variance in correctness and the system 0.6% (§3.3).

Table 1: Accuracy (%) at a matched budget of three calls (DIRECT uses one). Δ = DEBATE − VOTE in points, with paired bootstrap 95% CI.

Model	n	DIRECT	VOTE	CRITIQUE	DEBATE	Δ [95% CI]
GPT-OSS-120B	150	46.7	49.3	42.7	47.3	-2.0 [-8.7, +4.7]
Gemma 4 31B (no thinking)	150	48.7	48.7	47.3	57.3	+8.7 [+2.0, +15.3]
Flash-Lite	75	61.3	62.7	56.0	57.3	-5.3 [-13.3, +1.3]

- **Tools break the ceiling.** One tool-using call solves 47% of the questions that no other configuration solved, and more compute on top of tools adds comparatively little (§3.4).

2 Setup

Tasks. We draw 30 questions from each of five categories (150 in total): four from LAB-Bench [10] (protocol troubleshooting, sequence manipulation, database and literature questions; multiple choice) and open-ended sequence computations from LABBench2 SeqQA2 [11], scored with the official validators. All answers are scored deterministically, with no LLM judge.

Strategies. DIRECT makes one call. VOTE makes three independent calls and takes the majority answer. CRITIQUE solves, critiques its own solution, then revises. DEBATE has a molecular biologist and a skeptical reviewer solve independently, then an adjudicator decides. VOTE, CRITIQUE and DEBATE each use exactly three calls, and only visible outputs pass between calls (prompts in App. A).

Models. The matched-budget study uses GPT-OSS-120B [13] (low reasoning effort), Gemma 4 31B [17] (thinking disabled) and Gemini 3.5 Flash-Lite (low thinking) on all 150 questions (Flash-Lite on a fixed 75). To test reasoning effort and tools, we add GPT-6 Luna, GPT-6 Sol and GPT-5.6 Terra, and rerun GPT-OSS and Flash-Lite at high effort, on a fixed 30-question subset with 6 per category. Task IDs, prompts, decoding and the primary analysis were frozen before evaluation (App. B). All contrasts are paired over questions, with bootstrap 95% CIs and exact McNemar tests.

3 Results

3.1 Orchestration buys little at matched compute

Three calls spent on orchestration do no better than three calls spent on independent samples (Table 1). **Self-critique never beats voting** for any model (−6.7, −1.3, −6.7 points; pooled −4.5 [−9.3, +0.3]). It breaks correct answers as readily as it fixes wrong ones: relative to the vote it rescues 24%, 13% and 25% of errors but corrupts 38%, 16% and 26% of correct answers. The damage is worst on exact sequence manipulation (SeqQA: -23 and -20 points for GPT-OSS and Flash-Lite), where a fluent but wrong critique argues the model out of a correct computation. **Debate helps only the model that cannot think.** For Gemma, whose thinking is disabled, debate beats voting by +8.7 points ($p = 0.015$); for the two models that reason, it does not (−2.0 and −5.3).

3.2 Longer reasoning does not substitute for knowledge

Raising reasoning effort multiplies the cost without moving accuracy (Fig. 1). GPT-OSS at high effort generates $25\times$ more tokens and scores 47% against 50%; Flash-Lite spends $5\times$ more tokens and scores 67% against 70%. The one model that gains is GPT-6 Luna, which uses no reasoning tokens at low effort, and it gains only up to the level its low-effort debate had already reached (60% vs. 60%). Debate and reasoning are therefore substitutes, and neither gets past the plateau. Stacking orchestration on high effort adds nothing: GPT-6 Sol scores 70% with every strategy while the three-call strategies generate $\sim 3\times$ the tokens.

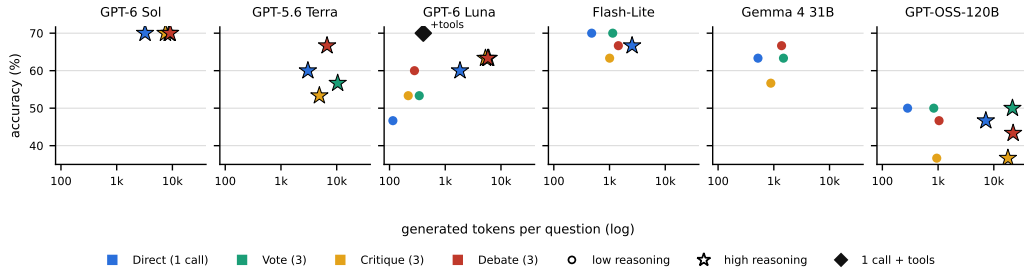


Figure 1: Every way of spending inference lands on the same per-model plateau; tools rise above it. Accuracy vs. generated tokens per question on a shared 30-question set. Colour marks the strategy; circles are low reasoning effort and stars high. Across nearly two orders of magnitude in tokens, accuracy barely moves within a model. The black diamond, one GPT-6 Luna call with web search and code execution, is the only configuration above its plateau, and it is among the cheapest.

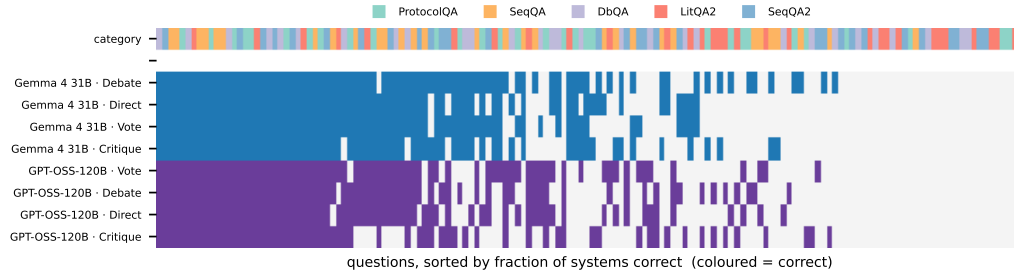


Figure 2: The question, not the system, decides. Correctness of every model $\times$ strategy system (rows) on every question (columns, sorted by the fraction of systems correct). All systems solve and fail the same questions. Question identity explains 56% [49, 62] of the variance in correctness; the system explains 0.6% [0.3, 1.9].

71 3.3 The question decides

72 A two-way decomposition of the correctness matrix attributes 56% of the variance to the question
 73 and 0.6% to the system (Fig. 2). The pattern holds across all 33 model, strategy and effort configu-
 74 rations on the shared subset, frontier models at high effort included (47% vs. 3.9%). Every system
 75 solves the same easy questions and fails the same hard ones. The 32 questions that no system
 76 solves are dominated by literature and database recall (18) and open-ended sequence computation
 77 (9). These are failures of knowledge, which no amount of deliberation can supply.

78 3.4 Tools break the ceiling

79 Giving GPT-6 Luna web search and code execution in a single call raises accuracy on the shared
 80 subset from 47% to 70% (+23 points [+7, +43], $p = 0.039$). That is more than any orchestration
 81 or reasoning setting achieves, at ~ 404 generated tokens per question. On the 32 questions that
 82 every orchestrated and reasoning-heavy system failed, the same single call reaches 47%, against 6%
 83 without tools (+41 points [+25, +56], $p = 0.0002$). Literature questions rise from 18% to 91% and
 84 database questions from 0% to 43%. Retrieval helping on these categories is expected, since they
 85 were designed for retrieval agents [10, 14]. The point is the contrast: no orchestration or reasoning
 86 setting moved these questions at all. Compute on top of tools adds comparatively little. High effort
 87 reaches 83% (+13, $p = 0.22$) and a three-call vote 77% (+7, $p = 0.73$), at $\sim 6\times$ and $\sim 3.5\times$ the
 88 tokens. Tool calls never reached the benchmark or its datasets.

89 4 Implications for building life-science agents

90 **Benchmark against a matched vote.** A multi-call agent should beat independent sampling with the
91 same number of calls, not a single call [9]. Most orchestration gains disappear under this baseline,
92 consistent with compute-matched studies of debate in general reasoning [4, 15, 18, 20] and with the
93 unreliability of intrinsic self-correction [7]. **Spend the budget on tools, not talk.** Critics, debaters
94 and longer chains of thought re-deliberate over knowledge the model does not have. Retrieval sup-
95 plies it, at a fraction of the tokens. **Build domain tools for computation.** A generic shell barely
96 helps open-ended sequence design and analysis (0% to 11%). The next gains lie in biology-specific
97 tool ecosystems and specialist models, not in more conversation between copies of the same model.

98 **Scope.** Our tasks have deterministic answers, and we study short-horizon question answering
99 rather than long-running discovery loops. The frontier and tool comparisons use a 30-question
100 subset, and the tool condition adds a one-sentence note announcing the tools.

101 5 Conclusion

102 On biology research tasks, how a model spends extra inference barely changes which questions it
103 gets right. This holds for voting, critique, debate and longer reasoning alike: the question explains
104 56% of the variance in correctness and the system 0.6%. The ceiling is knowledge, and a single
105 tool-using call breaks it where orchestration does not. Life-science agents should spend their budget
106 on tools, not talk.

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A Prompts

All prompts are single user messages. {QUESTION} is the question (with protocol, options or injected files) and {ANSWER INSTRUCTION} is the format line for multiple choice or the official SeqQA2 suffix.

Direct / Independent×3 / Critique call 1 (solve)

Solve the following biology research question.

{QUESTION}

{ANSWER INSTRUCTION}

Respond in exactly this format and keep it short:

<answer>YOUR ANSWER</answer>

CONFIDENCE: <probability between 0 and 1 that your answer is correct>

JUSTIFICATION: <at most two sentences>

189 **Critique call 2 (critique)**

190 Below are a biology research question and a proposed solution. Act as a critical reviewer: identify any
191 scientific, computational or reasoning error in the solution. If you think the answer is wrong, say which
192 answer you believe is correct and why. If you find no error, say so.

193
194 {QUESTION}

195
196 Proposed solution:
197 {CALL 1 OUTPUT}

198
199 Respond with at most four sentences, starting with "CRITIQUE:".

200 **Critique call 3 (revise)**

201 Below are a biology research question, a first attempt at solving it, and a reviewer's critique of that
202 attempt. Either may contain mistakes. Considering both, produce the final answer.

203
204 {QUESTION}

205
206 First attempt:
207 {CALL 1 OUTPUT}

208
209 Reviewer's critique:
210 {CALL 2 OUTPUT}

211
212 {ANSWER INSTRUCTION}

213
214 Respond in exactly this format and keep it short:
215 <answer>YOUR ANSWER</answer>
216 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
217 JUSTIFICATION: <at most two sentences>

218 **Debate call 1 (researcher)**

219 You are a molecular biology researcher. Solve the following biology research question independently.

220
221 {QUESTION}

222
223 {ANSWER INSTRUCTION}

224
225 Respond in exactly this format and keep it short:
226 <answer>YOUR ANSWER</answer>
227 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
228 JUSTIFICATION: <at most two sentences>

229 **Debate call 2 (skeptic)**

230 You are a skeptical scientific reviewer. Solve the following biology research question independently, looking
231 especially for subtle experimental, biological or computational errors and traps.

232
233 {QUESTION}

234
235 {ANSWER INSTRUCTION}

236
237 Respond in exactly this format and keep it short:
238 <answer>YOUR ANSWER</answer>
239 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
240 JUSTIFICATION: <at most two sentences>

241 **Debate call 3 (adjudicator)**

242 Two scientists independently analysed the biology research question below. Adjudicate between their analyses
243 (either or both may be wrong) and submit the final answer.

244
245 {QUESTION}

246

247 Analysis 1 (molecular biology researcher):
 248 {CALL 1 OUTPUT}
 249
 250 Analysis 2 (skeptical scientific reviewer):
 251 {CALL 2 OUTPUT}
 252
 253 {ANSWER INSTRUCTION}
 254
 255 Respond in exactly this format and keep it short:
 256 <answer>YOUR ANSWER</answer>
 257 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
 258 JUSTIFICATION: <at most two sentences>

259 B Pre-registration deviations

260 All task IDs, prompts, decoding settings and the primary analysis were frozen after a 10-question
 261 pilot and before any main-set call. Changes made after the freeze: (i) throttle and retry fixes for an
 262 undocumented 150 requests/hour provider limit (no completed call was re-run); (ii) 22 GPT-OSS
 263 questions served in full by a second provider of the same model (identical prompts, seeds and cap;
 264 excluding them leaves DEBATE – VOTE at -1.6), with partially served questions discarded and re-
 265 run on the primary provider; (iii) the reasoning-effort and tool arms on the fixed 30-question subset
 266 and the unsolved set, added after the freeze. Frontier models were served through their vendor CLI
 267 with web search, code execution and sub-agents disabled except in the tool condition (a first attempt
 268 with web search left on was discarded unanalysed); GPT-OSS at high effort was served by a second
 269 host whose low-effort accuracy matches the main run. The tool condition prepends one sentence
 270 announcing the tools, and all searches and commands were logged and checked for benchmark
 271 access (none).

272 C Additional results

Category	Gemma					Lite					OSS				
	D	I	C	M	Disagree	D	I	C	M	Disagree	D	I	C	M	Disagree
DbQA	56.7	53.3	40.0	56.7	33.3	40.0	40.0	40.0	33.3	26.7	46.7	46.7	43.3	36.7	36.7
LitQA2	33.3	30.0	33.3	36.7	46.7	60.0	53.3	53.3	46.7	26.7	20.0	30.0	40.0	36.7	43.3
ProtocolQA	56.7	56.7	60.0	70.0	20.0	66.7	80.0	86.7	73.3	26.7	56.7	50.0	43.3	46.7	33.3
SeqQA	56.7	56.7	66.7	73.3	46.7	73.3	73.3	53.3	66.7	13.3	63.3	66.7	43.3	60.0	33.3
SeqQA2	40.0	46.7	36.7	50.0	63.3	66.7	66.7	46.7	66.7	40.0	46.7	53.3	43.3	56.7	43.3

Table 2: Per-category accuracy (D=Direct, I=Independent×3, C=Critique, M=Multi). Numbers are percentages.